

ARCADENEXUS APP

Online Gaming Website
Product Requirement Document (PRD)

Project Title:	ArcadeNexus Online Gaming Website	Document Version:	v1.0.0 (Standard Format)
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1. PROJECT OVERVIEW

This is a custom online gaming website called ArcadeNexus. The website provides a modern, single-page retro-arcade gaming portal designed to run directly inside any modern web browser. It features a sleek, dark-themed responsive layout hosting three casual interactive client-side games (Tic-Tac-Toe, Snake, and Memory Match) rendered via modals, persisting user scores locally via the ``localStorage`` API.

1.1 Who each deliverable is for

Deliverable File	Who it is for
index.html (Semantic Markup)	Frontend integration developers and search engine indexers
styles.css (Responsive Styles)	Visual designers and developers maintaining responsive layouts
app.js & games.js (JS Engines)	System developers verifying local game loops and storage mechanisms
assets/ (Images & Audio folder)	Media designers updating textures, cards, or sounds
README.md (Quickstart Guide)	Reviewers and maintainers building or deploying the platform

1.2 Why the scope is shaped this way

This is a pure client-side vanilla stack application designed for instant load times and offline compatibility. No external runtime frameworks (e.g. React, Vue, jQuery) are used. User statistics persist session-free in ``localStorage`` without backend integration.

2. DELIVERABLES AT A GLANCE

#	Deliverable	Format	Exact file name
1	Main Layout Structure	HTML5 semantic file	<code>`index.html`</code>
2	Branding Styles & Layout	CSS3 stylesheet	<code>`styles.css`</code>
3	Core Portal Controller	ES6+ JavaScript	<code>`app.js`</code>
4	Mini-Games Engine Logic	ES6+ JavaScript	<code>`games.js`</code>
5	Project Documentation	Markdown document	<code>`README.md`</code>
6	Branded Vector Logo	SVG Image file	<code>`assets/images/logo.svg`</code>
7	Tic-Tac-Toe Cover Card	PNG Image file	<code>`assets/images/ttt_card.png`</code>
8	Snake Cover Card	PNG Image file	<code>`assets/images/snake_card.png`</code>
9	Memory Match Cover Card	PNG Image file	<code>`assets/images/memory_card.png`</code>
10	UI Selection Click Sound	MP3 Audio file	<code>`assets/audio/click.mp3`</code>
11	Point Scored Sound	MP3 Audio file	<code>`assets/audio/score.mp3`</code>
12	Game Over Ending Sound	MP3 Audio file	<code>`assets/audio/gameover.mp3`</code>
13	Loop Background Track	MP3 Audio file	<code>`assets/audio/cyber_beats.mp3`</code>

3. BRAND AND LOOK

ArcadeNexus uses a dark cybernetic retro design style. Developers must use these four colors only:





Typography: Orbitron for titles/headers/buttons; Inter for body copy and statistics tables.

4. THE PRODUCT AND THE STORY

The platform must get three core facts across: (1) ArcadeNexus provides a central web-based hub for quick retro games; (2) games operate fully client-side inside overlays; (3) user stats persist locally in the browser. The closing line in the footer is:
Play instantly, challenge your limits on ArcadeNexus.

4.1 The Five Section Layout Beats

Beat	Layout Section	What it does	Content Requirements
1	Header & Navbar	Branding & settings	Brand name in Orbitron, links to catalog, and mute toggle.
2	Hero Banner	Welcome & scroll scroll link	Neon-glowing slogan, quick intro, CTA button, and user welcome.
3	Games Catalog Grid	Selection interface	3 glassmorphic cards with cover assets, descriptions, and CTA.
4	Dashboard Section	Statistics visualizer	Overall metrics board, game-specific metrics, and reset data button.
5	Footer	Standardized closure	Licensing, copyright, author name, and brand closing line.

5. DELIVERABLE SPECIFICATIONS

- **index.html (Main Layout):** Full HTML5 semantics structure. References Google fonts and local style/script imports. Zero inline Javascript tags.
- **styles.css (Design Token Layout):** Root variables for color palette. Responsive grids/flexboxes. Media query breakpoints set at 576px and 992px.
- **app.js (Main Logic):** Manages navbar navigation, overlay modal loading, audio element initializations, and localStorage load/write operations.
- **games.js (Gameplay Logic):** Contains local event loops:
 - ***Tic-Tac-Toe:*** Interactive grid, single-player vs basic defense AI.
 - ***Snake:*** Canvas drawing context, speed incrementing 5% per 5 foods consumed, tail and edge collision loops.
 - ***Memory Match:*** Shuffled 4x4 board, turn count incrementing logic, 1-second flipped comparison delays.
- **README.md:** Developer quickstart manual, testing matrix, and licenses disclosure.

6. WEB QUALITY CHECKS

These checks define standard web layouts and must all pass:

1. **Direct DOM rendering:** No external JS libraries (jQuery, React) can be imported.
2. **Semantic layout:** Uses correct standard semantics structure (header, main, section, footer).
3. **Strict separation:** Event listeners must be bound dynamically in JS script files, not in HTML attributes.
4. **Zero errors:** Opening index.html directly from directory triggers no runtime or resource errors in Chrome.
5. **Responsive fluidity:** Under 576px wide, catalog shifts to single columns without horizontal viewport scrolling.

7. AUDIO SPECIFICATIONS

- **cyber_beats.mp3 (BGM):** Ambient loop sound track set at -20 LUFS. Plays loops automatically when toggled active.
- **UI SFX:** Click SFX (`click.mp3`), score updates (`score.mp3`), game over triggers (`gameover.mp3`) mixed at -14 LUFS.
- **Sound Settings Routing:** Audio elements must route through mute nodes controlled globally via the navbar mute toggle.
- **Licensing:** All assets must be commercial-friendly/Creative Commons, documented in README.md.

8. FILE NAMES AND FOLDER LAYOUT

```
ArcadeNexus/  
  index.html  
  styles.css  
  app.js  
  games.js  
  README.md  
  assets/  
    images/  
      logo.svg  
      ttt_card.png  
      snake_card.png  
      memory_card.png  
    audio/  
      click.mp3  
      score.mp3  
      gameover.mp3  
      cyber_beats.mp3
```

9. WHAT MUST BE TRUE TO ACCEPT

The delivery is accepted only when all of these are true:

- Unzipped folder is named exactly `ArcadeNexus` and has all files present.
- No off-palette colors show up on visual card surfaces, headers, or text.
- Game modes (TTT, Snake, Memory) load and play smoothly inside overlay modals.
- Stats dashboard persists values using localStorage under the key name `arcadenexus\_stats`.
- Muting sound toggle globally disables background music and click/score/gameover SFX.
- Every rubric item in rubrics.txt passes with verified local execution.